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United States Patent	12406795
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Teshigawara; Hidekazu et al.

Coil component and electronic device

Abstract

[PROBLEM] To suppress the formation of an external electrode extending to an unintended part.

[MEANS FOR RESOLUTION] A coil component **300** comprises: a magnetic substrate **10** including metal magnetic particles structured from a soft magnetic alloy that has iron as the main component; a coil conductor **30** provided on the magnetic substrate **10**; a glass film **50**, provided on a surface of the magnetic substrate **10**, made from a low-temperature sintered glass including a filler of titanium oxide, zirconium oxide, and/or tungsten oxide; and an external electrode **60**, provided in contact with the glass film **50**, connected electrically to the coil conductor **30**.

Inventors:	Teshigawara; Hidekazu (Tokyo, JP), Nagano; Masanori (Tokyo, JP)
Applicant:	TAIYO YUDEN CO., LTD. (Tokyo, JP)
Family ID:	83284063
Assignee:	TAIYO YUDEN CO., LTD. (Tokyo, JP)
Appl. No.:	17/667280
Filed:	February 08, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220301763 A1	Sep. 22, 2022

Foreign Application Priority Data

JP	2021-046344	Mar. 19, 2021
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Publication Classification

Int. Cl.: H01F27/28 (20060101); H01F1/147 (20060101); H01F27/255 (20060101); H01F27/29 (20060101)

U.S. Cl.:

CPC H01F27/29 (20130101); H01F1/147 (20130101); H01F27/255 (20130101); H01F27/2804 (20130101); H01F2027/2809 (20130101)

Field of Classification Search

USPC: 336/192; 336/198; 336/200; 336/232

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Primary Examiner: Ismail; Shawki S

Assistant Examiner: Hossain; Kazi S

Attorney, Agent or Firm: Stein IP, LLC

Background/Summary

FIELD OF TECHNOLOGY

(1) The present invention relates to coil components and electronic devices.

PRIOR ART

(2) Metal magnetic particles made from soft magnetic alloys with superior magnetic saturation characteristics are used instead of ferrite. For example, it is known that a coil component can be formed through layering a magnetic material layer, formed from a complex of an alloy powder that has, for example, iron, silicon, and aluminum as its main components, and a thermally, durable adhesive agent that includes borosilicate glass or silica, and an electrically conductive layer formed from a conductive powder (in, for example, Patent Document 1). Moreover, it is known that a coil component can be formed through layering a metal magnetic material layer formed from a metal magnetic material paste that includes metal magnetic particles and a thermally curable resin, and a conductor pattern formed from a conductive paste (in, for example, Patent Document 2).

Patent Documents

(3) [Patent Document 1] Japanese Unexamined Patent Application Publication H9-148118 [Patent Document 2] Japanese Unexamined Patent Application Publication 2007-27353

SUMMARY OF THE INVENTION

Problem Solved by the Present Invention

(4) When compared to ferrite, the electrical resistance of soft magnetic alloys is low. Because of this, when an external electrode is formed using an electrolytic plating method on the surface of a magnetic substrate that is formed using metal magnetic particles made from a soft magnetic alloy, in some cases the external electrode will be formed extending to unintended parts.

(5) The present invention was created in contemplation of the issue set forth above, and the object thereof is to prevent an external electrode from being formed extending to an unintended part.

Means for Solving the Problem

(6) The present invention is a coil component comprising: a magnetic substrate that includes a metal magnetic paste comprising a soft magnetic alloy that has iron as its main component; a coil conductor provided on the magnetic substrate; a glass film, provided on the surface of the magnetic substrate, comprising a low-temperature sintered glass that includes at least one filler selected from titanium oxide, zirconium oxide, and tungsten oxide; and an external electrode, provided in contact with the glass film, and connected electrically to the coil conductor.

(7) In this structure, the external electrode may be structured including a metal film and a plating film on the metal film.

(8) In this structure, the glass film may be structured including the filler at no less than 10 vol. %.

(9) In this structure, the glass film may be structured from a borosilicate glass that includes a titanium oxide tiller and/or a zirconium oxide tiller.

(10) In this structure, the glass film may be provided layered between the magnetic substrate and the external electrode, and may have external dimensions larger than those of the external electrode.

(11) In this structure, the glass film may be structured provided in contact with a ridge of the magnetic substrate.

(12) In this structure, the glass film may be structured having a color that is different from that of the external electrode.

(13) In this structure, the glass film may be given a color that is different from that of the external electrodes through inclusion of a filler made from manganese oxide, cobalt oxide, and/or ferrite.

(14) The present invention is an electronic device comprising a coil component set forth above and a circuit board on which the coil component is mounted.

Effects of the Invention

(15) The present invention enables prevention of formation of an external electrode extending to an unintended part.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

(1) FIG. 1 (a) is a perspective diagram of a coil component according to the first embodiment according to the present invention, and FIG. 1 (b) is a plan view wherein the coil component of FIG. 1 is viewed from the A direction.

(2) FIG. 2 is a cross-sectional diagram of the coil component of FIG. 1.

(3) FIG. 3 is an exploded perspective diagram of a magnetic substrate of the coil component of FIG. 1.

(4) FIG. 4 (a) is a bottom view of a coil component according to a first reference form, and FIG. 4

- (b) is a cross-sectional drawing in the vicinity of the external electrode.
- (5) FIG. 5 (a) is a cross-sectional drawing of the vicinity of an external electrode of a coil component according to a second reference form, and FIG. 5 (b) is a cross-sectional drawing prior to formation of the plating film.
- (6) FIG. 6 (a) is a schematic diagram of a scanning electron microscope image of the vicinity of the segregant, and FIG. 6 (b) through FIG. 6 (d) are schematic diagrams of energy dispersive x-ray analyses of the part in FIG. 6 (a).
- (7) FIG. 7 (a) is a schematic diagram of a scanning electron microscope image of a coil component according to a first embodiment, and FIG. 7 (b) through FIG. 7 (e) are schematic diagrams of energy dispersive x-ray analysis images of the part in FIG. 7 (a).
- (8) FIG. 8 (a) is a bottom view of a coil component according to a first embodiment, and FIG. 8 (b) is a cross-sectional drawing of the vicinity of the external electrode.
- (9) FIG. 9 is a cross-sectional drawing of a part of an external electrode provided on a rounded portion of a ridge of a magnetic substrate of a coil component after barrel polishing has been performed.
- (10) FIG. 10 is a plan view depicting the state of the glass film through the external electrode of the coil component according to a modified example of the first embodiment.
- (11) FIG. 11 (a) is a perspective diagram of a coil component according to a second embodiment according to the present application, and FIG. 11 (b) is a cross-sectional drawing of the coil component of FIG. 11 (a).
- (12) FIG. 12 (a) is a perspective diagram of a coil component according to a third embodiment according to the present invention, and FIG. 12 (b) is a cross-sectional drawing of the coil component of FIG. 12 (a).
- (13) FIG. 13 (a) is a perspective diagram of a coil component according to a fourth embodiment according to the present invention, and FIG. 13 (b) is a cross-sectional drawing of the coil component of FIG. 13 (a).
- (14) FIG. 14 is a cross-sectional drawing of a coil component according to a fifth embodiment according to the present invention.
- (15) FIG. 15 is a cross-sectional drawing of an electronic device equipped with a coil component according to the first embodiment according to the present invention.

MOST PREFERRED FORMS FOR CARRYING OUT THE PRESENT INVENTION

(16) Embodiments according to the present invention will be explained below, referencing the drawings as appropriate. However, this does not mean that the present invention is limited to the forms that are illustrated. Additionally, structural elements that are shared in a plurality of drawings are assigned identical reference symbols throughout the plurality of drawings. Be aware that, for convenience in explanation, not all drawings are necessarily drawn to scale.

First Embodiment

(17) FIG. 1 (a) is a perspective diagram of a coil component **300** according to a first embodiment according to the present invention, and FIG. 1 (b) is a plan view, viewed from the A direction, of the coil component **300** of FIG. 1. In FIG. 1 (b), the drawing is viewed through the external electrode **60**. FIG. 2 is a cross-sectional drawing of the coil component **300** of FIG. 1. FIG. 3 is an exploded perspective diagram of the magnetic substrate **10** of the coil component **300** of FIG. 1. Note that in FIG. 1 (a) and FIG. 2, for clarity in the drawings, the size of the glass film **50** at each of the pads on the four sides of the external electrodes **60** is different from that in FIG. 8, described below, where the exposed portion of the glass film **50** is illustrated enlarged. A layered inductor that is used as a passive element in a variety of circuits is depicted in FIGS. 1 (a), 1 (b), 2, and 3. The layered inductor is an example of a coil component that can be applied in the present invention. The present invention can be applied to power inductors used in DC-DC converters and to a variety of other components.

(18) The coil component **300** according to the first embodiment, as illustrated in FIGS. 1 (a), 1 (b),

2, and 3, comprises a magnetic substrate **10**, a coil conductor **30** that is embedded in the magnetic substrate **10**, a glass film **50**, and a pair of external electrodes **60** that are connected electrically to the coil conductor **30**. The “length” direction, “width” direction, and “height” direction of the coil component **300** are, respectively, the “L-axial” direction, the “W-axial” direction, and the “T-axial” direction. The L axis, W axis, and T axis are mutually, perpendicular. The coil axis A of the coil conductor **30** extends along the T-axial direction. While there is no particular limitation on the size of the magnetic substrate **10**, the length dimension (the dimension in the L-axial direction) may be, for example, between 0.2 mm and 6.0 mm, the width dimension (the dimension in the W-axial direction) may be, for example, between 0.1 mm and 4.5 mm, and the height dimension (the dimension in the T-axial direction) may be, for example, between 0.1 mm and 4.0 mm.

(19) The magnetic substrate **10** has a substantially rectangular prism shape. The outer surfaces of the magnetic substrate **10** are defined by six planes. Note that a “substantially rectangular prism shape” includes cases wherein the individual vertices are rounded, the individual edges (the boundary portions between sides) are rounded, and the individual surfaces are curved surfaces.

(20) In this first embodiment, when a reference is made to the “vertical direction” of the coil component **300**, this is in reference to the vertical direction in FIGS. 1 (a), 2, and 3. That is, the positive T-axial direction is up and the negative T-axial direction is down.

(21) The magnetic substrate **10** is formed including metal magnetic particles comprising a soft magnetic alloy that has iron as the main component. In each embodiment, the magnetic substrate **10** is formed through bonding together a plurality of metal magnetic particles through oxide films that are formed on the surfaces of the metal magnetic particles, but the magnetic substrate **10** may, alternatively, be formed through securing the plurality of metal magnetic particles using resin. The oxide film formed on the surface of a metal magnetic particle is an oxide of the soft magnetic alloy. Iron being the main component refers to a case wherein the proportion of iron in respect to the total amount of elements structuring the soft magnetic alloy is no less than 50 wt % (percent by weight), and may be 70 wt % or more, or may be 80 wt % or more, or may be 90 wt % or more.

(22) For example, the metal magnetic particles may be alloy particles that include iron and silicon, or may be alloy particles that include iron and a metal element M that is more easily ionized than iron. The metal element M may be, for example, chromium, aluminum, zirconium, titanium, manganese, or the like. As an example, the metal magnetic particles may be alloy particles of iron, silicon, and one or more types of metal elements M (for example, chromium and/or aluminum) that are more easily ionized than iron. The proportion of iron may be between 85 wt % and 97 wt %, the proportion of silicon may be between 1.5 wt % and 7 wt %, and the proportion of the metal element M may be between 1.5 wt % and 8 wt %. The metal magnetic particles may include unintentional impurities such as oxygen and/or carbon, or the like. The proportion of impurities may be 1 wt % or less. Additionally, the metal magnetic particles may include cobalt, nickel, copper, sulfur, phosphorus, boron, and the like. The composition ratios in the metal magnetic particles can be calculated through the ZAF method, for example, by imaging a cross section of the magnetic substrate **10** using a scanning electron microscope with a magnification between about 2000× and 20,000×, and performing energy dispersive x-ray analysis (EDS).

(23) The coil conductor **30** has a coil portion **32**. The coil portion **32** has coil patterns C11 through C15 and vias V1 through V4. The coil patterns C11 through C15 extend along the plane that is perpendicular to the coil axis A (the LW plane), and are mutually separated in the direction of the coil axis A (the T-axial direction). Of the coil patterns C11 through C15, coil patterns that are adjacent in the T-axial direction are connected electrically through the vias V1 through V4. The coil portion **32** that surrounds the coil axis A is formed thereby.

(24) One end of the coil portion **32** is connected electrically to one of the external electrodes **60** through a lead portion **34a** that is formed by vias V5 and V6. The other end of the coil portion **32** is connected electrically to the other external electrode **60** through a lead portion **34b** that is formed through vias V11 through V16.

(25) The magnetic substrate **10** has: a main unit portion **20** comprising the magnetic material layers **11** through **15** in which the coil portion **32** is provided; a top cover layer **16** comprising one or more layers, provided on the main unit portion **20**; and a bottom cover layer **17**, comprising one or more layers, provided below the main unit portion **20**. The coil patterns **C11** and vias **V1** and **V11** are formed in the magnetic material layer **11**, the coil pattern **C12** and the vias **V2** and **V12** are formed in the magnetic material layer **12**, the coil pattern **C13** and the vias **V3** and **V13** are formed in the magnetic material layer **13**, the coil pattern **C14** and the vias **V4** and **V14** are formed in the magnetic material layer **14**, and the coil pattern **C15** and the vias **V5** and **V15** are formed in the magnetic material layer **15**, Vias **V6** and **V16** are formed in the bottom cover layer **17**.

(26) The coil patterns **C11** to **C15** and vias **V1** through **V6** and **V11** through **V16** are formed from a metal material with superior conductivity, such as, for example, silver, palladium, copper, or aluminum, or from an alloy thereof.

(27) The glass film **50** is provided on the surface of the magnetic substrate **10**. In this first embodiment, the glass film **50** is provided, separated in the L-axial direction, on the bottom face, among the six bases that formed the outer surface of the magnetic substrate **10**. The glass film **50** is formed from a low-temperature sintered glass that includes a metal oxide filler of titanium oxide (TiO.sub.2), zirconium oxide (ZrO.sub.2), and/or tungsten oxide (WO.sub.3). For example, the glass film **50** is formed from a borosilicate glass that includes a titanium oxide filler. The inclusion proportion of the titanium oxide filler is no less than, for example, 10 vol % (percent by volume). The glass part and the metal oxide filler part in the glass film **50** can be identified through SEM observation. For example, the glass part will appear as irregular shapes, where the metal oxide filler part will appear as particles having grain boundaries. It is possible to discriminate between the two through contrast, and they can be distinguished from each other through carrying out component analysis through EDS analysis.

(28) The external electrode **60** is provided in contact with the glass film **50**. The external electrode **60** may be provided so as to overlap the glass film **50** in the plan view. In this first embodiment, the external electrode **60** is provided on the bottom face of the glass film **50**, except for the parts wherein the lead portion **34a** or lead portion **34b** is exposed from the magnetic substrate **10** and the parts for the opening **52a** or opening **52b** that is provided therearound. That is, except for the parts where the lead portion **34a** or lead portion **34b** is exposed from the magnetic substrate **10**, and the parts of the opening **52a** or the opening **52b** that is provided therearound, the glass film **50** is layered between the external electrode **60** and the magnetic substrate **10**. In the plan view, the glass film **50** is larger than the external electrode **60**. That is, in the plan view the external electrode **60** does not cover the entire glass film **50**, and there is a part where the glass film **50** can be seen alone. One of the pair of external electrodes **60** is filled into an opening **52a** that is provided in the glass film **50**, to connect electrically to the lead portion **34a**, and the other is filled into an opening **52b** that is provided in the glass film **50**, to connect electrically to the lead portion **34b**. The size of the opening **52a** or the opening **52b** is larger than the part wherein the lead portion **34a** or lead portion **34b** is exposed from the magnetic substrate **10**, and preferably is provided up to the periphery thereof, but instead may be provided with the same dimensions as the part wherein the lead portion **34a** or lead portion **34b** is exposed from the magnetic substrate **10**. Moreover, the size of the opening **52a** or opening **52b** may be smaller in relation to the parts wherein the lead portion **34a** or lead portion **34b** is exposed from the magnetic substrate **10**, and a portion of the parts wherein the lead portion **34a** or the lead portion **34b** is exposed from the magnetic substrate **10** may be covered with the glass film **50**. The external electrode **60** has a metal film **62** and a plating film **64** that is provided on the bottom face of the metal film **62**. The metal film **62** may be, for example, silver, palladium, copper, or an alloy thereof. The plating film **64** is a layered film of, for example, a nickel plating film and a tin plating film.

(29) [Method for Manufacturing]

(30) An example of a method for manufacturing the coil component **300** according to the first

embodiment will be explained. First a top layered body that serves as the top cover layer **16** is formed. The top layered body is formed through layering a plurality of magnetic material sheets. The magnetic material sheet is produced through, for example, coating a slurry onto the surface of a base film made from plastic, drying, and then cutting the slurry to a prescribed size after drying. The slurry is manufactured by mixing metal magnetic particles, made from a soft magnetic alloy that has iron as the main ingredient, with an organic binder and solvent, and the like. The organic binder uses, for example, a resin material with superior insulation, such as polyvinyl butyral (PVB) resin, epoxy resin, or the like. Toluene, for example, may be used for the solvent.

(31) A bottom layered body that will serve as the bottom cover layer **17** is formed next. The bottom layered body is formed through layering compound sheets wherein conductor vias of unbaked magnetic material sheets, described above, are provided. Through holes are formed at positions corresponding to vias **V6** and **V16** of the magnetic material sheet, and screen printing, for example, is used to fill the through holes with an electrically conductive paste. Note that the conductor vias may be formed through a method other than screen printing.

(32) Next an intermediate layered body that will serve as the main unit portion **20** is formed. The intermediate layered body is formed through layering, onto a magnetic material sheet that will serve as the magnetic material layers **11** through **15**, a compound sheet provided with an unbaked conductor pattern that will serve as the coil patterns **C11** through **C15**, and unbaked conductor vias that will serve as vias **V1** through **V5** and **V11** through **V15**. In order to form this compound sheet, first through holes are formed in positions, in the magnetic material sheet, described above, that will correspond to the vias **V1** through **V5** and **V11** through **V15**. The unbaked conductor pattern is then formed on the magnetic material sheet by printing an electrically conductive paste onto the magnetic material sheet using, for example, screen printing. In this case, the electrically conductive paste fills in the through holes formed in the magnetic material sheet. Through this, the unbaked conductor pattern that will serve as the coil patterns **C11** through **C15** and the unbaked conductor vias that will serve as vias **V1** through **V5** and **V11** through **V15** are formed on the magnetic material sheet. Note that the conductor patterns and the conductor vias may be formed through a method other than screen printing.

(33) Following this, an insulator pattern that will serve as the glass film **50** is formed on the bottom face of the magnetic material sheet that will serve as the bottommost layer of the magnetic substrate **10** (for example, the magnetic material sheet of the bottommost layer of the bottom layered body). The insulator pattern is formed through printing a glass paste made from a borosilicate glass that includes a titanium oxide filler. Next an unbaked conductor pattern that will serve as the metal film **62** for the external electrodes **60** is formed through coating an electrically conductive paste onto the bottom face of the insulator pattern that will serve as the glass film **50**. The insulator pattern that will serve as the glass film **50** and the unbaked conductor pattern that will serve as the metal film **62** and the external electrode **60** may be formed after forming the main unit layered body through contact bonding, described below, or after forming a chip layered body by dicing the main unit layered body, or formed after performing a heat treatment on the chip layered body, instead of the case wherein they are formed on the magnetic material sheet that will serve as the bottommost layer of the magnetic substrate **10**. In this case, at this stage the insulator pattern that will serve as the glass film **50** and the unbaked conductor pattern that will serve as the metal film **62** of the external electrodes **60** would not be formed on the magnetic material sheet that will serve as the bottommost layer of the magnetic substrate **10**.

(34) Next, the bottom layered body, the intermediate layered body, and the top layered body are layered together sequentially from the negative direction side to the positive direction side in the T-axial direction. The layered body is formed into the main unit layered body through thermal bonding using a pressing machine. In the step for forming the main unit layered body, pressure is applied, in the layering direction, to the insulator pattern that will serve as the glass film **50** and the conductor pattern that will serve as the metal film **62** of the external electrodes **60**. Because of this,

in some cases the insulator pattern that will serve as the glass film **50** and the conductor pattern that will serve as the metal film **62** of the external electrodes **60** may fill into the bottom face of the magnetic material sheet that will serve as the bottommost layer of the magnetic substrate **10**. When the insulator pattern that will serve as the glass film **50** and the unbaked conductor pattern that will serve as the metal film **62** of the external electrodes **60** are formed after contact bonding, at this stage the insulator patterns and conductor patterns are formed sequentially. The formation of each pattern can use any appropriate existing method. For example, screen printing may be used, or, for example, a transfer method may be used instead.

(35) Following this, a chip layered body is formed through dicing the main unit layered body into the desired size using a cutting machine such as a dicing machine or a laser machine tool. If the insulator pattern that will serve as a glass film **50** and the unbaked conductor pattern that will serve as the metal film **62** for the external electrodes **60** are to be formed after dicing, then, at this stage, the insulator pattern and the conductor pattern are formed sequentially. The formation of each pattern can use any appropriate existing method. For example, a printing method may be used, or, for example, a transfer method may be used, or, for example, a dipping method may be used. Following this, a heat treatment for baking is carried out on the chip layered body. The heat treatment is carried out at a prescribed temperature in an ambient gas that includes oxygen. Through this heat treatment, an oxide film is formed, comprising oxides of the material components of the metal magnetic particles, on the surfaces of the metal magnetic particles included in the bottom layered body, the intermediate layered body, and the top layered body, and the plurality of metal magnetic particles are bonded together through the oxide film. A polishing process, such as barrel polishing, or the like, is performed as necessary on the end portions of the chip layered body. If the insulator pattern that will serve as the glass film **50** and the conductor pattern that will serve as the metal film **62** of the external electrodes **60** is to be formed after the heat treatment, the insulator pattern and the conductor pattern are formed sequentially. The formation of each pattern may use any appropriate existing method. For example, screen printing may be used, or, for example, a transfer method may be used, or, for example, a dipping method may be used, or a thin film process, such as sputtering, or the like, may be used. After formation of the insulator pattern that will serve as the glass film **50** and the conductor pattern that will serve as the metal film **62** of the electrodes **60**, a heat treatment may be carried out, if necessary, to produce the glass film **50** and the metal film **62**.

(36) The external electrode **60**, which is made from a metal film **62** and a plating film **64**, is formed through forming a plating film, using an electrolytic plating method, on the surface of the conductor pattern that will serve as the metal film **62** of the external electrodes **60**. The coil component **300** is produced thereby.

(37) [First Reference Form]

(38) A coil component according to a first reference form differs from the coil component **300** according to the first embodiment in the point that no glass film is provided on the bottom face of the magnetic substrate. The other structures of the coil component according to the first reference form are identical to those of the coil component **300** according to the first embodiment. The coil component according to the first reference form is formed through a method that is identical to the method for manufacturing described in the first embodiment, except for the point that no glass film is formed.

(39) FIG. **4 (a)** is a bottom view of a coil component **1000** according to the first reference form, and FIG. **4 (b)** is a cross-sectional diagram of the vicinity of the external electrodes **160**, FIG. **4 (a)** is a diagram showing schematically a microscopic image wherein the bottom face of the coil component **1000** is viewed through an optical microscope, and FIG. **4 (b)** is a diagram showing schematically an SEM image wherein a cross section of the coil component **1000** is observed using a scanning electron microscope (SEM). In the coil component **1000** that is shown, the magnetic substrate **110** is formed through bonding metal magnetic particles, made from a soft magnetic

material of Fe—Si—Cr, through an oxide film that is formed on the surface of the metal magnetic particles. The coil conductor **130** is formed from silver, the external electrode **160** is formed, through an electrolytic plating method, from a metal film **162** made from silver, and the plating film **164** is formed from a nickel plating film **166** and a tin plating film **168**, formed through an electrolytic plating method.

(40) As depicted in FIGS. **4 (a)** and **4 (b)**, in the coil component **1000** according to the first comparative form, the plating film **164** extends further to the outside than the metal film **162**. In particular, the plating film **164** is formed extending greatly in the ridge vicinity (the edge vicinity) that structures the magnetic substrate **110**.

(41) Why the plating film **164** is formed extending further to the outside than the metal film **162** is thought to be due to the following reasons. The magnetic substrate **110** is formed through bonding together, through an oxide film formed on the surfaces of the metal magnetic particles, metal magnetic particles made from a soft magnetic alloy that has iron as the main component thereof. The surface of the magnetic substrate **110** receives physical impacts in the manufacturing process (for example, during barrel polishing, contamination removal processing, and the like), and thus it can be assumed that the oxide film that is formed on the surfaces of the metal magnetic particles becomes damaged. Because the electrical resistance of the soft magnetic alloy of the metal magnetic particles is low when compared to that of the oxide film that is produced on the surfaces thereof, when the oxide film on the surface of a metal magnetic particle becomes damaged, it can be assumed that the electrical resistance at the surface of the magnetic substrate **110** is low. In particular, because the vicinities of the ridges (the vicinities of the edges) in the magnetic substrate **110** and the vicinities of the vertices (the vicinities of the corners) are most likely to receive physical impacts in the manufacturing process, it can be assumed that the reductions in electrical resistance will be localized on only these parts. Given this, when forming the plating film **164** through an electrolytic plating method, it can be assumed that the plating film **164** will be formed extending further to the outside than the metal film **162**, and, in particular, it can be assumed that the plating film **164** will be formed extending further to the outside than the metal film **162** at those parts of the metal film **162** that are formed in the vicinities of the ridges (the vicinities of the edges) or in the vicinities of the vertices (the vicinities of corners) of the magnetic substrate **110**.

(42) Because, accompanying miniaturization of electronic devices, coil components are being made smaller as well, the spacing between the pair of external electrodes **160** has become narrower. Because of this, when the external electrode **160** is formed extending to parts that are not intended, the withstand voltage between the pair of external electrodes **160** will become inadequate, with the danger of producing faults in the coil component **1000**. In such a case it is possible to prevent such faults through the use of the present invention in particular.

(43) [Second Reference Form]

(44) In order to suppress the formation of external electrodes **160** extending to unintended parts, one may consider provision of an insulative glass film between the magnetic substrate **110** and the external electrodes **160**. Given this, in the coil component according to the second reference form a glass film made from a borosilicate glass (not including a titanium oxide filler) is provided between the magnetic substrate **110** and the metal film **162** for forming the external electrodes **160**. The component according to the second reference form is structured identically to that of the first embodiment except for the difference in the material for the glass film. Additionally, the coil component according to the second reference form is formed through the same method as the method for manufacturing, explained in the first embodiment, except for the difference of the material for the glass film.

(45) FIG. **5 (a)** is a cross-sectional drawing of the vicinity of an external electrode **160** of a coil component **1100** according to the second reference form, and FIG. **5 (b)** is a cross-sectional drawing prior to formation of the plating film **164**. FIG. **5 (a)** and FIG. **5 (b)** are diagrams showing schematically SEM images wherein the cross section of the coil component **1100** is observed using

a scanning electron microscope (SEM). In the coil component **1100** that is shown, the magnetic substrate **110** is formed through bonding metal magnetic particles, made from a soft magnetic material of Fe—Si—Cr, through an oxide film that is formed on the surface of the metal magnetic particles, the same as in the first reference form. The coil conductor **130** is formed from silver, the external electrode **160** is formed, through an electrolytic plating method, from a metal film **162** made from silver, and the plating film **164** is formed from a nickel plating film **166** and a tin plating film **168**, formed through an electrolytic plating method. The glass film **150** is formed from a borosilicate glass that does not include a titanium oxide filler.

(46) As depicted in FIG. 5 (a), even when the glass film **150** is formed on the surface of the magnetic substrate **110**, still the plating film **164** is formed extending further to the outside than the metal film **162**. Moreover, the region **180** of the magnetic substrate **110** in the periphery of the glass film **150** is observed to be different from the other regions of the magnetic substrate **110**.

Additionally, it was observed that segregant **190** has been formed between the glass film **150** and the nickel plating film **166**, between the magnetic substrate **110** and the nickel plating film **166**, etc. As depicted in FIG. 5 (b), the discolored region **180** and the segregant **190** were observed even prior to formation of the plating film **164**.

(47) The segregant **190** was evaluated next using energy dispersive x-ray analysis (EDS). FIG. 6 (a) is a schematic diagram of an SEM image wherein the vicinity of the segregant **190** was observed using a scanning electron microscope (SEM), where FIG. 6 (b) through FIG. 6 (d) are schematic diagrams of energy dispersive x-ray analysis (EDS) images of the part in FIG. 6 (a). In FIGS. 6 (b), 6 (c), and 6 (d), the K lines of iron, silicon, and oxygen, respectively, are mapped. The hatching region is the region wherein the applicable element is present at a high proportion, where a region where the hatching is dense indicates the inclusion proportion of the element being greater than a region wherein the hatching is sparse (where this applies to FIG. 7 (b) through FIG. 7 (e), below, as well).

(48) As depicted in FIG. 6 (a), in the image wherein energy dispersive x-ray analysis (EDS) has been performed, a glass film **150** is formed on the surface of the magnetic substrate **110**, and a segregant **190** is formed on the surface of the glass film **150**. As depicted in FIG. 6 (b) to FIG. 6 (d), it is confirmed that iron, silicon, and oxygen are distributed in the segregant **190**. It is assumed that the inclusion of iron, silicon, and oxygen in the segregant **190** is due to the following. Specifically, it is assumed that the glass film **150** softens during the heat treatment for forming the magnetic substrate **110**, and diffuses into the magnetic substrate **110** (which is assumed to be that which forms the discolored region **180**), and iron is accepted into the glass film **150** thereby, forming the segregant **190** that includes iron, silicon, and oxygen. It is assumed from the inclusion of iron, silicon, and oxygen in the segregant **190** that oxides of iron and oxides of silicon are included in the segregant **190**. While oxides of iron may be hematite (Fe.sub.2O.sub.3) or magnetite (Fe.sub.3O.sub.4), given that there is a large amount of oxygen in the segregant **190** it is assumed that a large amount of magnetite is included as the oxide of iron. The electrical resistivity of magnetite is low when compared to the resistivity of the silicon dioxide (SiO.sub.2) in the glass film **150**. Consequently, it is assumed that the electrical resistance is reduced through segregation of the segregant **190** on the bottom face of the glass film **150**.

(49) It is believed that the formation of the plating film **164** extending from the metal film **162** to the bottom face of the glass film **150** and the bottom face of the magnetic substrate **110** is because the electrical resistance is reduced through the segregation of the segregant **190** on the bottom face of the glass film **50** in this way.

Evaluation of the First Embodiment

(50) FIG. 7 (a) is a schematic diagram of an SEM image wherein a cross-section of the coil component **300** of the first embodiment was observed using a scanning electron microscope (SEM), where FIG. 7 (b) through FIG. 7 (e) are schematic diagrams of energy dispersive x-ray analysis (EDS) images of the part in FIG. 7 (a). In FIGS. 7 (b), 7 (c), 7 (d), and 7 (e), the K lines of

iron, silicon, oxygen, and titanium, respectively, are mapped. FIG. 7 (a) is a schematic diagram of an SEM image prior to formation of the plating film **64**, where the glass film **50** is formed on the surface of the magnetic substrate **10**, and a metal film **62** is formed on the surface of the glass film **50**. In the coil component **300** that was observed, the magnetic substrate **10** was formed through bonding together the metal magnetic particles, made of a soft metallic alloy of Fe—Si—Cr, through an oxide film formed on the surfaces of the metal magnetic particles. The metal film **62** of the external electrodes **60** was formed from silver, and the glass film **50** was formed from a borosilicate glass that includes a titanium oxide filler. In the example in FIG. 7 (a) through FIG. 7 (e), the inclusion proportion of the titanium oxide filler was 25 vol %.

(51) As depicted in FIG. 7 (a), in the coil component **300** according to the first embodiment, no discolored region was observed in the magnetic substrate **10**, and no segregant was observed on the surface of the magnetic substrate **10** or the surface of the glass film **50**. It is assumed that the reason why the use of a glass film **50** that includes a titanium oxide filler suppresses discoloration of the magnetic substrate **10** and formation of segregant on the surface of the magnetic substrate **10** and the surface of the glass film **50** is as follows. Specifically, it is assumed that the use, in the glass film **50**, of a borosilicate glass that includes a titanium oxide filler causes the glass film **50** to be resistant to softening, and, even if softened, lessens the reduction in viscosity of the glass film **50**, despite performing a heat treatment to form the magnetic substrate **10**. This is assumed to be because the glass component is held on the periphery of the titanium oxide filler. Because of this, it is assumed that, despite performing a heat treatment to form the magnetic substrate **10**, diffusion of the glass film **50** into the magnetic substrate **10** is prevented, where the result is suppression of discoloration of the magnetic substrate **10** and suppression of the formation of segregant on the surfaces of the magnetic substrate **10** and the glass film **50**.

(52) As depicted in FIG. 7 (b), iron was distributed within the magnetic substrate **10**, and substantially not distributed in the glass film **50**. As depicted in FIG. 7 (c), silicon was distributed in both the magnetic substrate **10** and the glass film **50**, but the glass film **50** had a larger inclusion proportion than in the magnetic substrate **10**. As depicted in FIG. 7 (d), a large amount of oxygen was included in the glass film **50**, and a small amount was included in the magnetic substrate **10** as well. As depicted in FIG. 7 (e), a large amount of titanium was included in the glass film **50**, and substantially not included in the magnetic substrate **10**.

(53) FIG. 8 (a) is a bottom view of a coil component **300** according to the first embodiment, and FIG. 8 (b) is a cross-sectional drawing of the vicinity of the external electrodes **60**. FIG. 8 (a) is a diagram showing schematically a microscopic image wherein the bottom face of the coil component **300** is viewed through an optical microscope, and FIG. 8 (b) is a diagram showing schematically an SEM image wherein a cross section of the coil component **300** is observed using a scanning electron microscope (SEM). In the coil component **300** that is shown, as with the above, the magnetic substrate **10** is formed through bonding metal magnetic particles, made from a soft magnetic material of Fe—Si—Cr, through an oxide film that is formed on the surface of the metal magnetic particles. The metal film **62** of the external electrodes **60** and the coil conductor **30** was formed from silver, and the glass film **50** was formed from a borosilicate glass that includes a titanium oxide filler. In the example in FIG. 8 (a) and FIG. 8 (b), the inclusion proportion of the titanium oxide filler was 25 vol %. The plating film **64** of the external electrodes **60** was formed through an electrolytic plating method, and was structured from a nickel plating film **66** and a tin plating film **68**.

(54) As depicted in FIG. 8 (a) and FIG. 8 (b), in the coil component **300** according to the first embodiment, formation of the plating film **64** extending from the metal film **62** to the outside was prevented. This is believed to be because the use of the glass film **50** that includes the titanium oxide filler prevented the formation of segregant at the surface of the glass film **50** and the surface of the magnetic substrate **10**, as depicted in FIG. 7 (a).

(55) In this way, in the first embodiment a glass film **50** made from a borosilicate glass that

includes a titanium oxide filler is provided on the surface of the magnetic substrate **10**, as illustrated in FIG. **1** (a) and FIG. **2**. The external electrode **60** is provided in contact with the glass film **50**. Through forming the glass film **50** from a borosilicate glass that includes the titanium oxide filler, formation of the segregant on the surface of the glass film **50** is suppressed, even when a heat treatment is carried out in the step for manufacturing the magnetic substrate **10**, preventing a reduction in electrical resistance. Thus the formation of the external electrodes **60** in contact with such a glass film **50** is able to suppress the external electrodes **60** from being formed extending to unintended parts.

(56) Note that while, in the first embodiment, set forth above, an example was illustrated wherein the glass film **50** was formed from a borosilicate glass that includes a titanium oxide filler, there is no limitation thereto, but rather it may instead be formed from a borosilicate glass that includes a filler that is an oxide of zirconium (zirconium oxide), as a metal element that has little reactivity with Fe and that has a high softening point when made into a glass. Moreover, for the same reasons as described above, the glass film **50** may instead be made from a borosilicate glass that includes a tungsten oxide filler. In addition, the glass film **50** is not limited to being a borosilicate glass, but only need be a low-temperature sintered glass. The formation of the segregant on the surfaces of the magnetic substrate **10** and the glass film **50** is suppressed in these cases as well.

(57) Moreover, in the first embodiment the external electrode **60** is structured including a metal film **62** and a plating film **64** on the metal film **62**. While there is a tendency for the plating film **64** to be formed extending outward when the electrical resistance on the surface of the glass film **50** is low, the use of the glass film **50** made from a low-temperature sintered glass that includes a filler that is titanium oxide, zirconium oxide, and/or tungsten oxide suppresses the reduction in electrical resistance on the surface of the glass film **50**, making it possible to prevent the plating film **64** from being formed extending outward.

(58) Moreover, in the first embodiment the glass film **50** includes the filler that is titanium oxide, zirconium oxide, and/or tungsten oxide at no less than 10 vol % (percent by volume). The formation of the external electrodes **60** extending out to an unintended part can be suppressed effectively thereby. From the perspective of suppressing the formation of the external electrodes **60** extending outward to unintended parts, preferably the inclusion proportion of the filler is no less than 15 vol %, more preferably no less than 20 vol %, and even more preferably no less than 25 vol %. On the other hand, from the perspective of manufacturing the glass paste for forming the glass film **50**, if the inclusion proportion of the filler were too great it would increase the viscosity of the glass paste, and the dispersibility would be poor. Because of this, if the inclusion proportion of the filler were not less than 60 vol %, manufacturing of the glass paste in industry, would be difficult, where no greater than 50 vol % is preferred, and, in consideration of the viscosity characteristics when considering ease of printing, no greater than 40 vol % is even more preferred. The glass part and the filler part in the glass film **50** can be identified through SEM observations. For example, the glass part is observed in irregular shapes. The filler part is observed as particles having grain boundaries. The two can be distinguished through contrast, and can be distinguished through carrying out component analysis through EDS analysis. The inclusion proportion of the filler within the glass film **50** can be calculated from the area of the filler and the area of the other glass parts, in a plan view, when the glass film **50** is observed enlarged to between 2000× and 20,000× by an SEM. The filler part and the other glass parts can be distinguished through contrast, and the respective areas can be calculated using, for example, binarization or gradation technologies. Moreover, even when distinguishing by contrast is difficult, the respective areas can be calculated by distinction between the filler part and the other glass parts by finding the element compositions of those parts through carrying out mapping through EDS analysis.

(59) Moreover, in the first embodiment, the glass film **50** is in contact with the ridges (edges) of the magnetic substrate **10**. The ridges (edges) and vertices (corners) of the magnetic substrate **10** receive physical impacts through the manufacturing process (for example, barrel polishing,

processing to remove contamination, and the like), and thus tend to have reduced electrical resistance through damage to the oxide film that formed on the surfaces of the metal magnetic particles, so there is a tendency for the external electrodes **60** to be formed extending outward. Consequently, the provision of the glass film **50** in contact with the ridges (edges) of the magnetic substrate **10** enables effective suppression of the formation of the external electrodes **60** extending to unintended parts.

(60) Moreover, in the first embodiment, the glass film **50** is layered between the magnetic substrate **10** and the external electrode **60**, except for the parts wherein the lead portion **34a** or lead portion **34h** is exposed from the magnetic substrate **10** and the parts of the opening **52a** and the opening **52b** that are provided in the periphery thereof, and has an outer shape that is larger than that of the external electrode **60**. That is, in the plan view the external electrode **60** does not cover all of the glass film **50**, and the glass film **50** is a part that can be seen alone. This enables effective suppression of formation of the external electrodes **60** extending out to unintended parts.

(61) Moreover, in the first embodiment preferably the glass film **50** is of a color that is different from that of the external electrode **60**. This enables the determination as to whether or not the external electrode **60** is formed extending out to unintended parts to be done through carrying out image recognition. In other words, it is possible to prevent, in image recognition, mistaking the region wherein the glass film **50** is provided as being a region formed by the extension of the external electrodes **60**. The difference in color from that of the external electrodes **60** may be applied through the glass film **50** including a filler that is manganese oxide, cobalt oxide, and/or ferrite. Having the glass film **50** be a color that is different from that of the external electrodes **60** can be achieved easily thereby.

(62) Note that in the first embodiment, set forth above, preferably barrel polishing is carried out on the chip layered body after heat treatment in the manufacturing process for the coil component **300**. Carrying out barrel polishing chamfers the ridges (edges) of the magnetic substrate **10**, facilitating exposure of the glass film **50**, as depicted in FIG. **9**. Although the ridges (edges) of the magnetic substrate **10** are locations wherein there is a tendency for the plating film **64** to be formed extending outward, exposing these locations to the glass film **50** enables suppression of the extension of the plating film **64**, enabling suppression of formation of the external electrodes **60** to unintended parts.

Modified Example of the First Embodiment

(63) FIG. **10** is a plan view of a coil component **310** according to a modified example according to the first embodiment. In FIG. **10**, the view is through the external electrode **60**. With the coil component **310** according to the modified example of the first embodiment, as depicted in FIG. **10**, the glass film **50** has a squared-C shape that extends along the ridge (edge) of the magnetic substrate **10**. The ridge (edge) of the magnetic substrate **10** is a location at which there is a tendency for the plating film to be formed extending out. Even when the glass film **50** is in this shape, the extension of the plating film is actually prevented at locations where there is a tendency for the plating film to be formed extending out, making it possible to suppress formation of the external electrodes **60** at unintended parts.

Second Embodiment

(64) FIG. **11 (a)** is a perspective diagram of a coil component **400** according to a second embodiment according to the present invention, and FIG. **11 (b)** is a cross-sectional drawing of the coil component **400** in FIG. **11 (a)**. As depicted in FIG. **11 (a)** and FIG. **11 (b)**, in the coil component **400** according to the second embodiment, the glass film **50** is provided over the entire region of the bottom face of the magnetic substrate **10** with the exception of the parts wherein the lead portion **34a** and the lead portion **34b** are exposed from the magnetic substrate **10** and the parts of the opening **52a** and the opening **52b** that are provided therearound. The other structures are identical to the coil component **300** according to the first embodiment, and thus explanations thereof are omitted.

(65) In the second embodiment as well, a glass film **50**, formed from a low-temperature sintered glass that includes a metal oxide filler of titanium oxide (TiO.sub.2), zirconium oxide (ZrO.sub.2), and/or tungsten oxide (WO.sub.3) is provided, on the surface of the magnetic substrate **10**. For example, a glass film **50** made from a borosilicate glass that includes, for example, a titanium oxide filler is provided, and the external electrode **60** is provided in contact with the glass film **50**. The formation of the external electrodes **60** extending to an unintended part is suppressed thereby, in the same manner as with the first embodiment.

(66) Moreover, in the second embodiment the provision of the glass film **50** over the bottom face of the magnetic substrate **10** as a whole enables a further suppression of formation of the external electrodes **60** in unintended parts. Moreover, the provision of the glass film **50** over the bottom face of the magnetic substrate **10** as a whole means that the glass film **50** will exist between the external electrodes **60**, enabling suppression of shorting between the external electrodes **60**.

Third Embodiment

(67) FIG. **12 (a)** is a perspective diagram of a coil component **500** according to a third embodiment according to the present invention, and FIG. **12 (b)** is a cross-sectional drawing of the coil component **500** in FIG. **12 (a)**. In the coil component **500** according to the third embodiment, as depicted in FIG. **12 (a)** and FIG. **12 (b)**, the glass film **50** extends from the bottom face of the magnetic substrate **10** onto the end faces (the WT faces). The external electrode **60** also, similarly, extends from the bottom face of the magnetic substrate **10** onto the end faces. The lead portions **34a** and **34b** are formed extending in the L-axial direction, and connect to the external electrodes **60** on the end faces of the magnetic substrate **10**. To form the glass film **50** and the external electrodes **60** on the end faces (the WT faces) of the magnetic substrate **10**, preferably they are formed in a step after dicing, and thus the formation of the insulator pattern that will serve as the glass film **50** and of the unbaked conductor pattern that will serve as the metal film **62** of the external electrodes **60** is through sequentially forming individually in a stage after dicing or after the heat treatment. The other structures are identical to those of the coil component **300** according to the first embodiment, and thus explanations thereof will be omitted.

(68) In the third embodiment as well, a glass film **50**, formed from a low-temperature sintered glass that includes a metal oxide filler of titanium oxide (TiO.sub.2), zirconium oxide (ZrO.sub.2), and/or tungsten oxide (WO.sub.3) is provided, on the surface of the magnetic substrate **10**. For example, a glass film **50** made from a borosilicate glass that includes, for example, a titanium oxide filler is provided, and the external electrode **60** is provided in contact with the glass film **50**. The formation of the external electrodes **60** extending to an unintended part is suppressed thereby, in the same manner as with the first embodiment. The external electrodes **60** extending from the bottom face of the magnetic substrate **10** onto the end faces enables the end faces (the WT faces) to be secured by solder frit when mounting the coil component onto a circuit board. This makes it possible to achieve high mounting strength.

Fourth Embodiment

(69) FIG. **13 (a)** is a perspective diagram of a coil component **600** according to a fourth embodiment according to the present invention, and FIG. **13 (b)** is a cross-sectional drawing of the coil component **600** in FIG. **13 (a)**. In the coil component **600** according to the fourth embodiment, as depicted in FIG. **13 (a)** and FIG. **13 (b)**, the glass film **50** extends from the bottom face of the magnetic substrate **10** to the top face via the end faces (the WT faces) and the side faces (the LT faces). The external electrodes **60** also, similarly, extend from the bottom face of the magnetic substrate **10** to the top face, via the end faces and the side faces. The lead portions **34a** and **34b** are formed extending in the L-axial direction, and connect to the external electrodes **60** on the end faces of the magnetic substrate **10**. To form the glass film **50** and the external electrodes **60** on the end faces (the WT faces) and the side faces (the LT faces) of the magnetic substrate **10**, preferably they are formed in a step after dicing, and thus the formation of the insulator pattern that will serve as the glass film **50** and of the unbaked conductor pattern that will serve as the metal film **62** of the

external electrodes **60** is through sequentially forming individually in a stage after dicing or after the heat treatment. The other structures are identical to the coil component **300** according to the first embodiment, and thus explanations thereof are omitted.

(70) In the fourth embodiment as well, a glass film **50**, formed from a low-temperature sintered glass that includes a metal oxide filler of titanium oxide (TiO.sub.2), zirconium oxide (ZrO.sub.2), and/or tungsten oxide (WO.sub.3) is provided, on the surface of the magnetic substrate **10**. For example, a glass film **50** made from a borosilicate glass that includes, for example, a titanium oxide filler is provided, and the external electrode **60** is provided in contact with the glass film **50**. The formation of the external electrodes **60** extending to an unintended part is suppressed thereby, in the same manner as with the first embodiment. The external electrode **60** extending from the bottom face of the magnetic substrate **10** to the top face via the end faces (the WT faces) and the side faces (the LT faces) enables the end faces (the WT faces) and the side faces (the LT faces) to be secured by solder frit when mounting the coil component onto a circuit board. This makes it possible to achieve high mounting strength. Moreover, this enables both the top face and the bottom face to be used as the face for mounting onto the substrate, rendering it unnecessary to lineup the up/down orientations during packaging.

Fifth Embodiment

(71) In the first embodiment through the fourth embodiment, described above, examples were presented of layered coil components, but in the fifth embodiment an example of a winding-type coil component will be presented. FIG. **14** is a cross-sectional diagram of a coil component **700** according to a fifth embodiment according to the present invention. As depicted in FIG. **14**, in the coil component **700** according to the fifth embodiment, the magnetic substrate **10** has a shaft portion **70**, and the pair of flange portions **72** that are provided on both ends of the shaft portions **70**. The coil conductor **30** is formed through wrapping, around the shaft portion **70**, a lead wire **74** that has an insulating coating. Glass films **50** are provided on both of the pair of flange portions **72**, and external electrodes **60** are provided on the glass films **50**. The external electrode **60** that is provided on one of the flange portions **72** is connected electrically to one end of the coil conductor **30**, and the external electrode **60** that is provided on the other flange portion **72** is connected electrically to the other end of the coil conductor **30**. Note that while in FIG. **14** an example is shown wherein the glass films **50** and external electrodes **60** are provided on both of the two flange portions **72**, the glass film **50** and two external electrodes **60** may instead be provided on one of the two flange portions **72**.

(72) In the fifth embodiment as well, a glass film **50**, formed from a low-temperature sintered glass that includes a metal oxide filler of titanium oxide (TiO.sub.2), zirconium oxide (ZrO.sub.2), and/or tungsten oxide (WO.sub.3) is provided, on the surface of the magnetic substrate **10**. For example, a glass film **50** made from a borosilicate glass that includes, for example, a titanium oxide filler is provided, and the external electrode **60** is provided in contact with the glass film **50**. The formation of the external electrodes **60** extending to an unintended part is suppressed thereby, in the same manner as with the first embodiment.

(73) FIG. **15** is a cross-sectional drawing of an electronic device **800** comprising a coil component **300** according to the first embodiment according to the present invention. As depicted in FIG. **15**, in the electronic device **800** the coil component **300** is mounted on a circuit board **90**. In the coil component **300**, the external electrodes **60** are bonded to land electrodes **94** of the circuit board **90** by solder **92**, to mount onto the circuit board **90**. Because, in the coil component **300**, formation of the external electrodes **60** on unintended parts is suppressed, this enables prevention of adverse effects on other electronic components, even when other electronic components are packaged near the coil component **300**. The electronic device **800** may be, for example, a smart phone, a tablet terminal, a game console, or the like. Note that the coil component mounted on the circuit board **90** may instead be a coil component **400** through **700** according to the second embodiment through the fifth embodiment.

EMBODIMENTS

(74) Although the invention according to the present application has been explained in detail through embodiments and reference examples, this does not mean that the invention according to the present application is limited to the aspects described in the embodiments. In the coil components of the embodiments and reference examples below, the length dimensions (dimensions in the L-axial direction) are 2.0 mm, the width dimensions (the dimensions in the W-axial direction) are 1.6 mm, and the height dimensions (the dimensions in the T-axial direction) are 0.65 mm. Moreover, in the coil components in the embodiments and reference examples, the form is one wherein the glass film is provided separated in two in the L-axial direction, as in the coil component according to the first embodiment and first reference form, described above. The dimension in the L-axial direction of each of the glass films is 0.5 mm, the dimension in the W-axial direction is 1.6 mm, and the dimension in the T-axial direction (the thickness) is 20 μm . For the metal film for structuring each of the external electrodes, the dimension in the L-axial direction is 0.4 mm, the dimension in the W-axial direction is 1.52 mm, and the dimension in the T-axial direction (the thickness) is 20 μm , where the ridge (edge) sides of the magnetic substrate have spacing of $40\ \mu\text{m} \pm 20\ \mu\text{m}$ each in respect to the edges of the glass film, and for the sides facing each of the external electrodes, the spacing is $100\ \mu\text{m} \pm 20\ \mu\text{m}$ in respect to the edges of the glass film. The T-axial direction dimensions (the thicknesses) of the plating film for structuring the external electrodes are 1 μm for Ni plating and 3 μm for Sn plating.

Embodiment 1

(75) The coil component of Embodiment 1 was manufactured through the process set forth below. For the raw material particles, an alloy pattern with a compositional ratio of silicone: 3.5 wt %, chromium: 1.5 wt %, with the remaining portion being iron and inevitable impurities was used, and a slurry was prepared including this alloy powder, toluene (a solvent) and polyvinyl butyral (a binder). The slurry was coated onto the surface of a base film through a doctor blade method and dried with a hot air dryer, where, after drying, the slurry was cut to a prescribed size to manufacture a magnetic material sheet. If necessary, through holes were formed at prescribed positions of the magnetic material sheet, followed by printing, onto the surface of the magnetic material sheet, an electrically conductive paste including silver powder, butyl carbitol (a solvent), and ethyl cellulose (a binder), followed by drying with a hot air dryer, to manufacture a magnetic material sheet having a precursor for conductor patterns and vias.

(76) A glass paste made from a borosilicate glass that includes a titanium oxide filler was printed onto the bottom face of the magnetic material sheet that would serve as the bottommost layer of the magnetic substrate, and dried using a hot dryer, to form an insulator pattern that is the precursor of the glass film. Following this, an electrically conductive paste including silver powder, butyl carbitol (a solvent), and ethyl cellulose (a binder) was printed onto the insulator pattern and dried by a hot air dryer, to form a conductor pattern that is a precursor to the metal film for forming the external electrodes.

(77) After layering these magnetic material sheets in a prescribed sequence, they were press-bonded together at a prescribed pressure. The press-bonded unit following press-bonding was cut into chip units and for carrying out a process to remove the binder, and the like, a heat treatment was carried out for one hour at 800° C. in an ambient gas that includes oxygen. The inclusion proportion of titanium oxide filler included in the glass film after the heat treatment was 10 vol %. Following this, an electrolytic plating method was used to form a plating film, made from a nickel plating film and a tin plating film, on the surface of the conductor pattern that is the precursor for the metal film for structuring the external electrodes, to form the external electrodes that are structured from the metal film and the plating film.

Embodiment 2

(78) A coil component was manufactured through the same method as in the first embodiment, except for the inclusion proportion of the titanium oxide filler included in the glass film after the

heat treatment being 12 vol %.

Embodiment 3

(79) A coil component was manufactured through the same method as in the first embodiment, except for the inclusion proportion of the titanium oxide filler included in the glass film after the heat treatment being 15 vol %.

Embodiment 4

(80) A coil component was manufactured through the same method as in the first embodiment, except for the inclusion proportion of the titanium oxide filler included in the glass film after the heat treatment being 20 vol %.

Embodiment 5

(81) A coil component was manufactured through the same method as in the first embodiment, except for the inclusion proportion of the titanium oxide filler included in the glass film after the heat treatment being 25 vol %.

Embodiment 6

(82) A coil component was manufactured through the same method as in the first embodiment, except for the inclusion proportion of the titanium oxide filler included in the glass film after the heat treatment being 40 vol %.

Embodiment 7

(83) A coil component was manufactured through the same method as in the first embodiment, except for the inclusion proportion of the titanium oxide filler included in the glass film after the heat treatment being 50 vol %.

Embodiment 8

(84) A coil component was manufactured through the same method as in the first embodiment, except for the inclusion proportion of the titanium oxide filler included in the glass film after the heat treatment being 60 vol %.

Embodiment 9

(85) A coil component was manufactured using the same manufacturing method as in Embodiment 1 except for forming the glass film using a glass paste made from a borosilicate glass that includes a zirconium oxide filler, with the inclusion proportion of the zirconium oxide filler included in the glass film after the heat treatment being 25 vol %.

Embodiment 10

(86) A coil component was manufactured using the same manufacturing method as in Embodiment 1 except for forming the glass film using a glass paste made from a borosilicate glass that includes a zirconium oxide filler, with the inclusion proportion of the zirconium oxide filler included in the glass film after the heat treatment being 50 vol %.

Reference Example

(87) A coil component was manufactured using the same method as in Embodiment 1, except for forming the insulator pattern that is the precursor for the glass film using a glass paste made from a borosilicate glass that does not include a titanium oxide filler or a zirconium oxide filler.

(88) Energy dispersive x-ray analyses (EDS) were performed on magnetic substrate cross-sections in the vicinity of the external electrodes of the coil components of Embodiment 1 through Embodiment 10 and the Reference Example, and the distances between the tip ends of the metal films (silver films) that structure the external electrodes and the tip ends of the tin plating films that are included in the plating films were measured. Given this, if the distance between the metal film and the tin plating film was 20 μm or greater, this was defined as the plating extending. This evaluation was carried out for 20 coil components each for Embodiment 1 through Embodiment 5 and the reference example, and the proportions of coil components wherein plating extension occurred were calculated.

(89) The results obtained are given in Table 1.

(90) TABLE-US-00001 TABLE 1 Filler Inclusion Proportion [vol %] Rate of Occurrence of

TiO.sub.2 ZrO.sub.2 Plating Extension [%] Embodiment 1 10 — 30 Embodiment 2 12 — 15
Embodiment 3 15 — 0 Embodiment 4 20 — 0 Embodiment 5 25 — 0 Embodiment 6 40 — 0
Embodiment 7 50 — 0 Embodiment 8 60 — 0 Embodiment 9 — 25 0 Embodiment 10 — 50 0
Reference Example — — 100

(91) As in Table 1, plating extension occurred at a rate of 100% (in all 20 of the coil components) in the reference example that used a glass film that did not include a titanium oxide filler or a zirconium oxide filler. On the other hand, in Embodiment 1, which used a glass film that included a titanium oxide filler at 10 vol %, the proportion of coil components wherein plating extension occurred was reduced to 30%, and in Embodiment 2, which used a glass film that included 12 vol % titanium oxide filler, the proportion of coil components wherein plating extension occurred was reduced to 15%. Moreover, no plating extension occurred in any of the 20 coils of Embodiment 3 through Embodiment 8, which used glass films that included titanium oxide filler that 15 vol %, 20 vol %, 25 vol %, 40 vol %, 50 vol %, or 60 vol %, nor in Embodiment 9 or Embodiment 10, which used glass films that included zirconium oxide filler at 25 vol % or 50 vol %.

(92) In this way, it was confirmed that the use of a glass film that includes a titanium oxide filler or a zirconium oxide tiller suppresses the occurrence of plating extension, enabling prevention of formation of the external electrodes extending to unintended parts. Moreover, it was confirmed that the inclusion proportion of the titanium oxide filler or zirconium oxide filler preferably is no less than 10 vol %, where no less than 12 vol % is more preferred, and no less than 15 vol % is even more preferred.

(93) While embodiments of the invention according to the present application were described in detail, the invention according to the present application is not limited to these specific embodiments, and various modifications and changes are possible within the range of the spirit and intent of the present application, described in the patent claims.

EXPLANATIONS OF REFERENCE SYMBOLS

(94) **10**: Magnetic Substrate **11** through **15**: Magnetic Material Layers **16**: Top Cover Layer **17**: Bottom Cover Layer **20**: Main Unit Portion **30**: Coil Conductor **32**: Coil Portion **34a**, **34b**: Lead Portions **50**: Glass Film **52a**, **52b**: Openings **60**: External Electrode **62**: Metal Film **64**: Plating Film **70**: Shaft Portion **72**: Flange Portion **74**: Lead Wire **90**: Circuit Board **92**: Solder **94**: Land Electrode **110**: Magnetic Substrate **130**: Coil Conductor **150**: Glass Film **160**: External Electrode **162**: Metal **164**: Plating Film **166**: Nickel Plating Film **168**: Tin Plating Film **180**: Region **190**: Segregant **300**, **310**, **400**, **500**, **600**, **700**, **1000**, **1100**: Coil Components **800**: Electronic Device

Claims

1. A coil component comprising: a magnetic substrate including metal magnetic particles structured from a soft magnetic alloy that has iron as the main component; a coil conductor provided on the magnetic substrate; a glass film, provided on a surface of the magnetic substrate, made from a low-temperature sintered glass including a filler of titanium oxide, zirconium oxide, and/or tungsten oxide; and an external electrode, provided in contact with the glass film, connected electrically to the coil conductor, wherein the external electrode includes a metal film, and a plating film on the metal film, and wherein an edge of the external electrode is located on the glass film and is in contact with the glass film.

2. The coil component set forth in claim 1, wherein: the glass film includes the filler at no less than 10 vol %.

3. The coil component set forth in claim 1, wherein: the glass film is made from a borosilicate glass that includes a titanium oxide filler and/or a zirconium oxide filler.

4. The coil component set forth in claim 1, wherein: the glass film is provided layered between the magnetic substrate and the external electrode, and has an external shape that is larger than that of the external electrode.

5. The coil component set forth in claim 1, wherein: the glass film is provided in contact with a ridge of the magnetic substrate.
 6. The coil component set forth in claim 1, wherein: the glass film has a color that is different from that of the external electrode.
 7. The coil component set forth in claim 6, wherein: the glass film has a color that is different from that of the external electrode through inclusion of a filler made from manganese oxide, cobalt oxide, and/or ferrite.
 8. An electronic device comprising: the coil component set forth in claim 1; and a circuit board on which is mounted the coil component.
 9. The coil component set forth in claim 1, wherein: the coil conductor comprises a coil portion; a lead portion electrically connects an end of the coil portion to the external electrode; and the glass film encircles a region of contact of the lead portions and the bottom of the magnetic substrate.
 10. The coil component set forth in claim 1, wherein: the metal film fills an opening of the glass film and the metal film covers the glass film.
 11. The coil component set forth in claim 1, wherein: the glass film coves multiple surfaces of the magnetic substrate.
 12. The coil component set forth in claim 9, wherein: the lead portion extends along a length of the magnetic substrate.
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